What survives implementation costs?

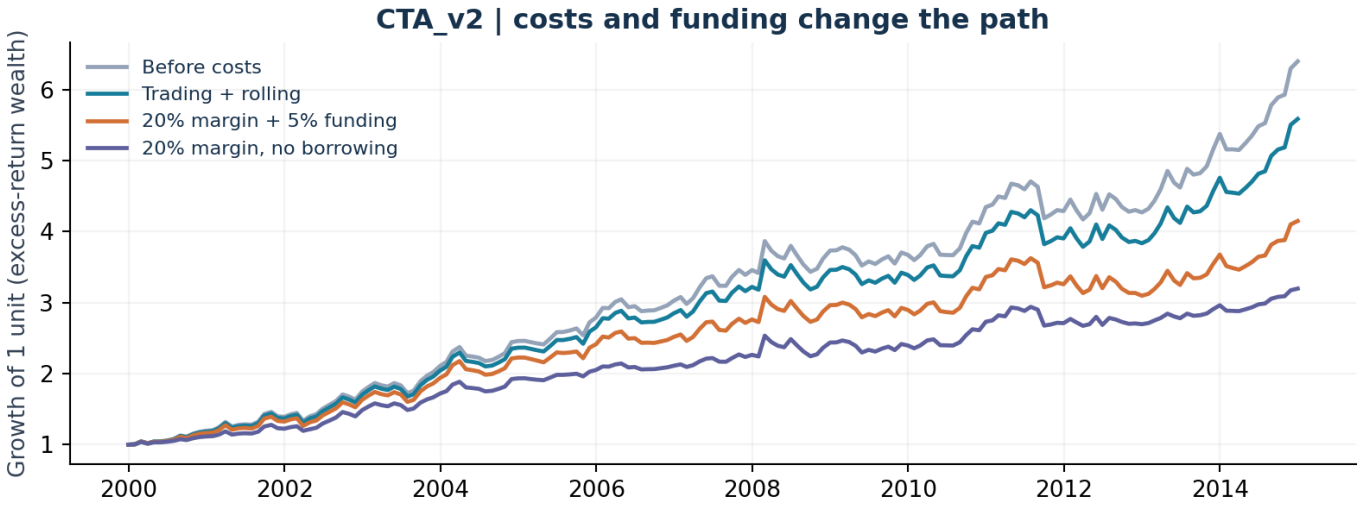
ORIE 5260 | Managed futures | 2000-2014 historical holdout (180 months)

CTA_v2: Sharpe 1.172 before costs, 1.092 after trading and rolling, and 0.915 under the 20% margin / 5% incremental funding scenario. Funding is charged on the cash shortfall, not gross notional.

1 / Baseline comparison

Strategy	Gross SR	Net SR: T+R	Net SR: +funding	T+R net mean / yr
STATIC	0.723	0.698	0.673	10.75%
TSMOM_12	0.839	0.796	0.766	10.20%
VT11_historical	0.959	0.906	0.844	11.45%
VT11_ewma	1.177	1.101	0.932	12.11%
CTA_v1	1.161	1.075	0.865	11.85%
CTA_v2	1.172	1.092	0.915	12.13%

T+R: 2 bps per unit traded plus 2 bps per combined roll, four rolls/year. Funding column adds 20% uniform margin, 20% cash reserve and 5% incremental financing. Annual means are arithmetic annualizations; returns exclude cash income.



The no-borrow path changes positions to fit cash capacity; its performance is not a pure fee deduction. All paths start at 1 on 31 December 1999. Lines summarize a previously inspected historical period.

Design: six original strategies reproduced; no signal or risk-model parameter retuning. Training ends in 1999. Zero-cost full-execution returns and turnover match the legacy engine.

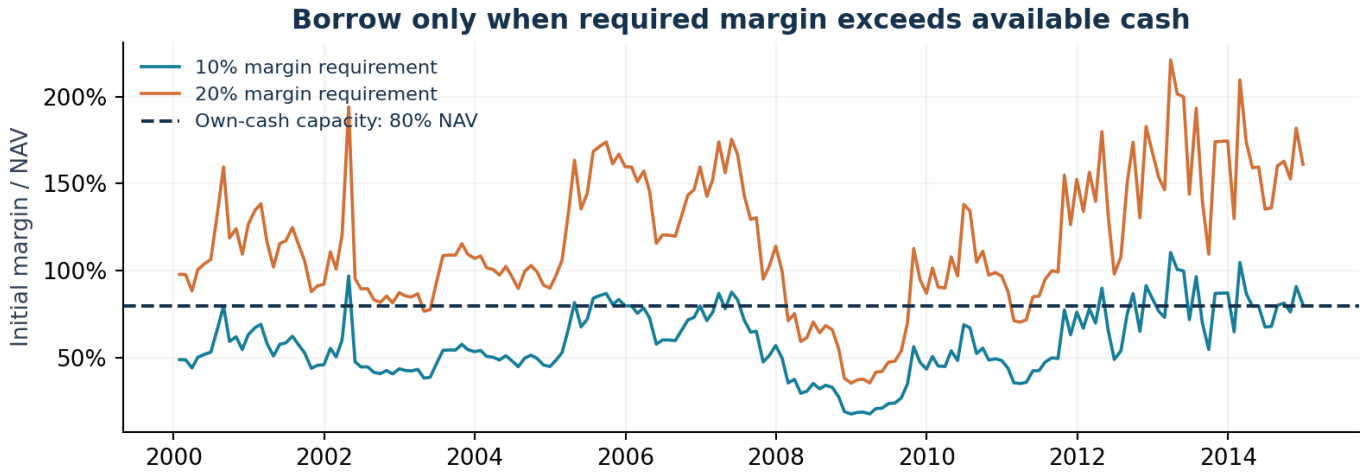
2 / Margin and financing

Notional exposure is not the borrowed amount. Margin is posted collateral, not a return expense.

Monthly model
Margin / NAV = margin rate x gross notional / NAV
Borrowing / NAV = max(margin / NAV - 80%, 0)
Financing cost / NAV = borrowing / NAV x annual incremental rate / 12

CTA_v2 scenario	Avg margin / NAV	Avg debt / NAV	Months with debt	Funding / yr	Net SR
10% margin / 2% rate	58.5%	1.4%	27/180	0.03%	1.090
20% margin / 5% rate	117.0%	40.1%	155/180	2.00%	0.915
30% margin / 8% rate	175.5%	96.5%	171/180	7.72%	0.400
20% / no borrowing	76.9%	0.0%	0/180	0.00%	0.947

All scenarios use a 20% NAV cash reserve and base trading/rolling costs. Rates are hypothetical incremental borrowing spreads net of collateral/reference cash income, not observed all-in loan quotes.



Reading the chart: at 10% margin, borrowing is needed in 27 of 180 months. At 20%, that rises to 155 months; peak debt is 141.0% of NAV. Averages alone conceal funding pressure.

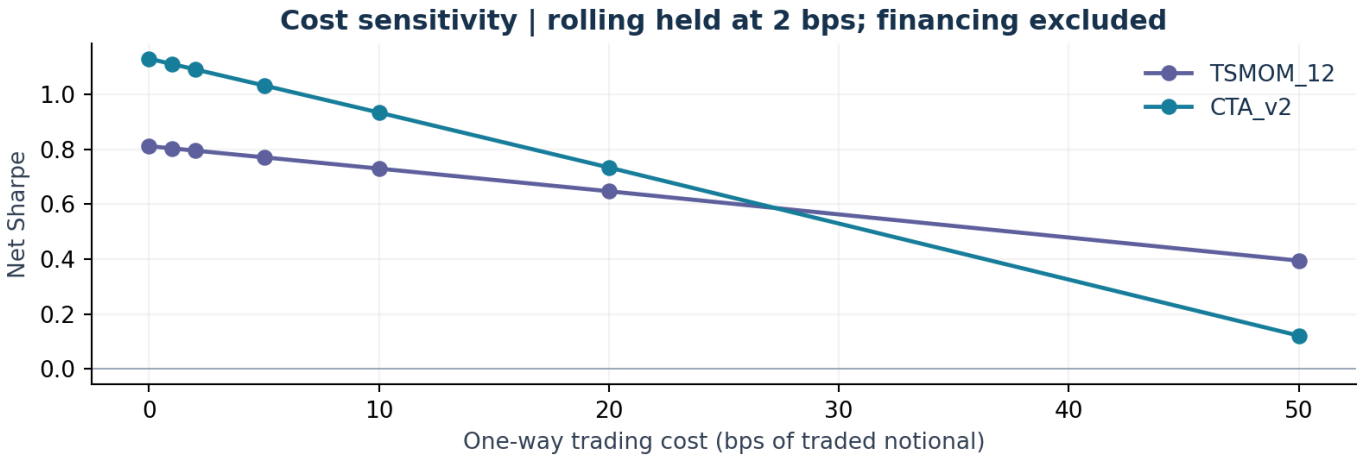
No-credit alternative: scale exposure whenever margin would exceed 80% NAV. This binds in 155 months; realized volatility becomes 8.59% and net annual mean return 8.14%. Recomputed turnover includes the scaling trades.

Scope: monthly initial-margin proxy; credit assumed available in borrow cases. No intramonth variation margin, offsets, haircuts, lender limits or forced intraday liquidation. Zero modeled borrowing does not mean no margin-call risk.

3 / Execution choice and next steps

Policy	Train net SR	Holdout net SR	Turnover / yr	Max drawdown
full	1.392	1.092	22.32x	-11.97%
buffer_0.05	1.390	1.093	21.95x	-11.99%
buffer_0.10	1.386	1.096	21.28x	-11.86%
buffer_0.20	1.376	1.112	19.61x	-11.55%
partial_0.25	1.228	1.096	14.08x	-12.49%
partial_0.50	1.316	1.101	16.58x	-11.83%
partial_0.75	1.365	1.096	19.22x	-11.76%

Training selected **full**. Policy results include trading/rolling costs but exclude financing, isolating the original execution experiment. Holdout improvements are diagnostic, not a basis for changing the selected rule. Entries, exits and reversals execute immediately.



Grid points are scenarios, not a calibrated impact model. CTA_v2's Sharpe advantage over TSMOM_12 disappears between 20 and 50 bps one-way trading cost. Funding is excluded in this comparison.

Priorities for the next iteration

1. Calibrate per-contract margin, liquidity, roll calendars and funding terms.
2. Add daily variation-margin stress, credit limits and final-position concentration checks.
3. Validate execution rules with walk-forward selection and new data; then optimize net returns subject to capital constraints.

Sources: project execution-study CSVs and manifest; Lecture 8, pp. 24, 25 and 32. The 2000-2014 period was already examined in the midterm. Estimates are not proof of significance or production readiness. Reproduce: run_execution_study.py, then build_results_summary.py.